

Math 550 – Winter 2024
Assignment 3

Question 1

Intuition is that each iteration of forming $S(X)$ "fills in" the space between points of X , and that after a certain number of iterations, there are no gaps left in any direction in the space, resulting in a convex set. Intuition leads us to try to show that $S_{\lceil \log_2 d+1 \rceil}(X) = \text{conv}(X)$.

Claim 1. S_k corresponds to convex combinations of at most 2^k points of X .

The proof follows immediately from the claim since any $x \in \text{conv}(X)$ is a convex combination of at most $d+1$ points of X (Carathéodory), thus letting $k = \lceil \log_2(d+1) \rceil$, S_k covers all convex combinations of at most $2^{\lceil \log_2(d+1) \rceil} = d+1$ points of X , thus $\text{conv}(X) \subseteq S_{\lceil \log_2(d+1) \rceil}(X)$. Conversely, $S_k(X)$ for any k contains only convex combinations of points in X , thus $S_{\lceil \log_2(d+1) \rceil}(X) \subseteq \text{conv}(X)$. Since $S(X)$ contains convex combinations of 2 points of X , $S_2(X)$ contains convex combinations of 4 points of X , and so on.

Proof of claim. Base case is simple, consider $k = 1$, clear that $S(X)$ contains all convex combinations of 2 points of X , because $S(X)$ contains all line segments between pairs of points in X . The induction step is as follows, suppose x is a convex combination of at most 2^k points of X , then

$$\exists \lambda_1, \dots, \lambda_{2^k} \text{ such that } \sum_{i=1}^{2^k} \lambda_i = 1 \text{ and } \lambda_i \geq 0 \forall i, \text{ and}$$

$$\begin{aligned} x &= \lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_{2^k} x_{2^k} \\ &= (\lambda_1 x_1 + \dots + \lambda_{2^{k-1}} x_{2^{k-1}}) + (\lambda_{2^{k-1}+1} x_{2^{k-1}+1} + \dots + \lambda_{2^k} x_{2^k}) \end{aligned}$$

$$\text{letting } A = \sum_{i=1}^{2^{k-1}} \lambda_i \text{ and } B = \sum_{i=2^{k-1}+1}^{2^k} \lambda_i \text{ then}$$

$$x = A \cdot \overbrace{\left(\frac{\lambda_1}{A} x_1 + \dots + \frac{\lambda_{2^{k-1}}}{A} x_{2^{k-1}} \right)}^{\in S_{k-1}(X)} + B \cdot \overbrace{\left(\frac{\lambda_{2^{k-1}+1}}{B} x_{2^{k-1}+1} + \dots + \frac{\lambda_{2^k}}{B} x_{2^k} \right)}^{\in S_{k-1}(X)}$$

where $A + B = 1$, , thus $x \in S_k(X)$.

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Question 2

(a)

$x, y, z \in \mathbb{R}^2$ form a triangle, where the vertices have pairwise distance at most 1. Any triangle such triangle can be embedded inside a equilateral triangle, call it Δ with all vertices having pairwise distance 1 (shrink the edges of Δ accordingly). Therefore, it's enough to show it for Δ . Taking the circumcenter of Δ , call it m , the unique point equidistant to all three vertices. Since it's found by taking the point at which all perpendicular bisectors of Δ (60 degree angles) intersect, then the distance from m to any other vertex is calculated as such

$$\begin{aligned} \sqrt{\left(\frac{\tan 30}{2}\right)^2 + \left(\frac{1}{2}\right)^2} &= \sqrt{\left(\frac{\sqrt{3}}{3 \cdot 2}\right)^2 + \left(\frac{1}{2}\right)^2} \\ &= \sqrt{\frac{3}{36} + \frac{1}{4}} \\ &= \sqrt{\frac{1}{3}} \\ &= \frac{1}{\sqrt{3}} \end{aligned}$$

Thus $\Delta \subseteq \overline{B(m, \frac{1}{\sqrt{3}})}$ the closed disc centered at m with radius $\frac{1}{\sqrt{3}}$.

(b)

Let C_x be the disc centered at x of radius $\frac{1}{\sqrt{3}}$. We know that for any $\{x, y, z\} \subseteq X$, $C_x \cap C_y \cap C_z \neq \emptyset$. Indeed, by (a), for any three pts $\{x, y, z\} \subseteq X$, there exists some point m such that $\|m - x\|, \|m - y\|, \|m - z\| \leq \frac{1}{\sqrt{3}}$, thus $m \in C_x \cap C_y \cap C_z$. By Helly's, since $\{x, y, z\}$ is arbitrary, then $\bigcap_{x \in X} C_x \neq \emptyset$, thus there exists some $m \in \mathbb{R}^2$ such that $\|m - x\| \leq \frac{1}{\sqrt{3}}$ for any $x \in X$, and the result follows.

(c)

We first do the case with 4 points, where any 4 points in $\mathbb{R}^3$ form a hyperplane in $\mathbb{R}^2$. Consider the extremal case first, which is the regular tetrahedron with edge length 1. It has a circumcircle of radius $\sqrt{\frac{8}{3}}$, thus using the the sam argument as (b), every finite set with diameter at most 1 can be contained in a closed ball of radius $\sqrt{\frac{8}{3}}$. Then the goal is to show that $\forall \varepsilon > 0$, then finding a counter example for $c = \sqrt{\frac{8}{3}} - \varepsilon$ which I don't know how to do, I've never taken a course in discrete geometry before.

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(a)

To do

(b)

To do

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4

(a)

We want to use fractional Helly's theorem with $\alpha = \frac{1}{4}$, want to show that there exists at least $\alpha \binom{n}{3} = \frac{\binom{n}{3}}{4}$ sets I where $|I| = 3$ where $\bigcap_{i \in I} C_i \neq \emptyset$ to be able to conclude that there exists some X such that $|X| \geq n\beta = \frac{n}{12}$ where $\beta = \frac{\alpha}{3}$ such that $\bigcap_{i \in X} C_i \neq \emptyset$. We know that there are $\binom{n}{4}$ ways of forming a bundle of 4 sets, each of which contains a triplet that share a common element, by assumption. But each triplet is contained in $n - 3$ bundles of 4 sets. That means, there are $\frac{\binom{n}{4}}{n - 3}$ sets I where $|I| = 3$ such that $\bigcap_{i \in I} C_i \neq \emptyset$. But observe that

$$\begin{aligned} \frac{\binom{n}{4}}{n - 3} &= \frac{n!}{(n - 4)!4!(n - 3)} \\ &= \frac{n!}{(n - 3)!3!} \cdot \frac{1}{4} \\ &= \frac{\binom{n}{3}}{4} \end{aligned}$$

(b)

We want to use fractional Helly's theorem again. We want to get a lower bound on the number of I 's such that $|I| = d + 1$ such that $\bigcap_{i \in I} C_i \neq \emptyset$. We know that every bundle of p sets that we choose contains a sub-bundle of $d + 1$ sets with a common element, there are $\binom{n}{p}$ ways of choosing such a bundle, but every sub-bundle of $d + 1$ sets is contained in $\binom{n - (d + 1)}{p - (d + 1)}$ bundles of p sets, therefore there exists at least the following amount of I 's with the property:

$$\begin{aligned} \binom{n}{p} \cdot \frac{1}{\binom{n - (d + 1)}{p - (d + 1)}} &= \frac{n!}{p!(n - p)!} \cdot \frac{(p - (d + 1))!(n - p)!}{(n - (d + 1))!} \\ &= \frac{n!}{(d + 1)!(n - (d + 1))!} \cdot \frac{(p - (d + 1))!(d + 1)!}{p!} \\ &= \binom{n}{d + 1} \cdot \frac{1}{\binom{p}{d + 1}} \end{aligned}$$

Could've gotten the same result knowing that

$$\binom{n}{p} \binom{p}{d + 1} = \binom{n}{d + 1} \binom{n - (d + 1)}{p - (d + 1)}$$

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By double counting pairs of (bundle, sub-bundle) The result follows immediately by fractional Helly's letting $\alpha = \frac{1}{\binom{p}{d+1}}$ and $c = \beta = \frac{\alpha}{d+1}$

(c)

This is probably done by following the proof for fractional Helly's, let's try it out.